End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech., Semester: 1st
(December, 2025)

Subject Code: BAI 104	**Subject: Programming Fundamentals**
Time :3 Hours	**Maximum Marks :60**

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8 =20)**	**CO Mapping**
	a) Compare an algorithm and a program in Python with suitable examples.		CO1
	b) List the advantages of object oriented programming.		CO1
	c) What is the need of bitwise operators in Python?		CO2
	d) Write a Python program that accepts a tuple and sorts it in descending order.		CO2
	e) List the difference between lists and tuples with example code.		CO3
	f) Differentiate between default and keyword argument in function call with suitable example code.		CO3
	g) Demonstrate the importance of base case in recursion with example code.		CO4
	h) Write a python program to find transpose of a 2D matrix using numpy.		CO4
	UNIT I		**CO1**
Q2	Describe the concept with suitable example code of (a)Abstraction (b) Inheritance	(5+5)	
Q3	(a) Write a program to print the given pattern. 1 1 2 1 2 3 1 2 3 4 1 2 3 4 5 (b) Write a Python program that accepts a person's age and prints whether they are **a minor, an adult, or a senior citizen.**	(5+5)	
	UNIT II		**CO2**
Q4	(a) Write a program that takes two lists and returns True if they have at least one common member. (b) Write a Python program to delete a given key-value pair from a dictionary and display the updated dictionary.	(5+5)	

Q5	Write a Python program to multiply two matrices. Moreover, handle the condition where matrix multiplication is not possible due to incompatible dimensions.	(10)	
	UNIT III		**CO3**
Q6	(a) Write a program to check whether a given word is present in a statement using function. If the word is found, display a suitable message; otherwise, indicate that it is not present. (b) What are **command line arguments** in Python? Explain their purpose and usage with a suitable example program.	(6+4)	
Q7	(a) Write a python program to generate a 5-digit CAPTCHA using random number generator and functions. (b) Create a NumPy array containing numbers from 1 to 10. Calculate its mean, median, and standard deviation.	(5+5)	
	UNIT IV		**CO4**
Q8	Write a Python program to append content to a text file and handle the exception that occurs when trying to open a non-existing file.	(5+5)	
Q9	Write a Python program to a) Generates 1000 random numbers from a standard normal distribution. b) Plots a histogram with 30 bins, sky-blue bars, and black edges. c) Adds X and Y axis labels and a title. d) Displays the histogram plot.	(10)	